

Chongkun Deng

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chongkun-deng.github.io

Research Interests

Statistical inference and learning for structured data; high-dimensional probability; geometric and topological methods; ordinal comparisons, graph structure, and optimization.

Education

Columbia University, Graduate School of Arts and Sciences

September 2023 - December 2024

M.A. in Statistics; GPA: 3.82/4.00

University of California San Diego

September 2019 - June 2023

B.S. in Mathematics; GPA: 3.69/4.00; Major GPA: 3.67/4.00

Research Manuscripts

1. Farthest-cell triplet entropy: high-dimensional shell limits and hyperbolic curvature amplification.

arXiv:2609.02362, September 2026. Preprint.

Defines conditional entropy for the farthest-prototype label given random triples. Derives high-dimensional limits for isotropic Euclidean and hyperbolic radial models and a curvature amplification factor. Studies model-calibrated, scale-invariant curvature inference, the external length scale needed for absolute curvature, and sensitivity to angular anisotropy.

2. Tree Recognition and Graph Edge Minimization from Anchored Distance Comparisons.

Manuscript, September 2026.

Studies positive weighted tree and graph realizations of complete anchored distance rankings on observed vertices. Recovers candidate tree topology in quadratic time, then tests weights with a sparse LP in root distances. Gives integer weight bounds and infeasibility certificates. For graphs, proves incompatible edge omissions, a linear gap, and the sharp worst-case minimum edge count $n(n-1)/2 - 1$.

Research Experience

Independent Research

December 2024 - present

Statistical and geometric structure from ordinal observations

- Develop mathematical results for random geometry, tree recognition, and graph realization when observations supply comparisons rather than numerical distances.
- Combine probabilistic limit theory, graph algorithms, linear programming, simulations, and exact computational verification.

RTG Reading Group, University of California San Diego

March 2022 - June 2022

Algebraic curves and discrete valuation rings

- Studied and presented mathematical material in a collaborative reading group guided by graduate students.

Teaching Experience

Teaching Assistant, University of California San Diego

September 2020 - March 2021

MATH 20C: Calculus and Analytic Geometry for Science and Engineering

- Led discussion sections and office hours and graded assignments and examinations over two quarters.
- Prepared course materials and contributed to discussions about course design.

Honors and Competitions

Harvard-MIT Mathematics Tournament, General Round

November 2018

Top 3%, rank 24.

Technical Skills and Languages

Programming: Python, R

Languages: Chinese (native), English (fluent), Japanese (conversational)